

Soap Film (Minimal Surface)

Non-planar wire: discrete Laplace relaxation on z; iridescent mesh — Plateau sketch.

Wire: flat bottom + vertical sides at z=0 + sinusoidal top · Interior: discrete Laplace.

FRAME & MESH

Top edge wave amplitude

1.28 (sim)

Relaxation steps / frame

14

Grid resolution

50

MEASURED VALUES

Grid N

50

About this model

A soap film stretched across a non-planar wire frame forms a minimal surface, a shape of the smallest possible area for its given boundary. This simulator visualizes that surface through a computational physics technique known as Laplace relaxation. The film is modeled as a discrete mesh of points, each with a height (z-coordinate). The core principle is that at equilibrium, the mean curvature at any point on the film is zero, a condition derived from minimizing the surface area under the constraint of the fixed boundary wire. The simulator approximates this by iteratively adjusting the height of each interior mesh point to the average height of its immediate neighbors, solving the discrete Laplace equation: $\nabla^2 z = 0$. This relaxation process continues until the mesh stabilizes, revealing the smooth, saddle-shaped minimal surface. The iridescent colors are not modeled with full wave optics but are represented schematically using interference-based

coloring to evoke the thin-film interference patterns seen in real soap films, which depend on local film thickness. Key simplifications include treating the soap film as having negligible thickness and uniform surface tension, ignoring gravity and dynamic effects like film drainage. By interacting with this simulator, students learn how complex physical shapes emerge from simple local averaging rules, connect the mathematical concept of a harmonic function (solution to Laplace's equation) to a tangible physical system, and observe the direct link between area minimization and zero mean curvature.

Who it's for: Undergraduate students in physics, mathematics, or engineering studying calculus of variations, partial differential equations, or introductory continuum mechanics.

KEY TERMS

- Minimal Surface
- Laplace's Equation
- Mean Curvature
- Surface Tension
- Plateau's Laws
- Thin-Film Interference
- Harmonic Function
- Relaxation Method

How it works

A **thin soap film** minimizes **area**; for **small slopes** the height **$z(x,y)$** over a plane obeys **$\nabla^2 z = 0$** with **Dirichlet** data on the **wire** — the same **Laplace** equation as this **Jacobi relaxation** on a square grid. The **frame** here is schematic: **flat bottom**, **vertical sides** at **$z = 0$** , and a **sinusoidal top** so the boundary is **not planar**; the interior **smooths** into a **harmonic** surface (pedagogical **stand-in** for a full **Plateau** minimal surface).

Key equations

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Discrete Laplace: z_ij <- average of four orthogonal neighbors (Jacobi)
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Frequently asked questions

Why does the soap film always form that smooth, saddle-shaped surface?

The film minimizes its total surface area to reduce its potential energy, which is proportional to area due to surface tension. A surface with zero mean curvature, like a saddle shape, achieves this local area minimization. This is a direct consequence of the physical principle that systems evolve to their lowest energy state.

The simulator uses an averaging process. What physics law does that represent?

The averaging rule—setting a point's height to the average of its neighbors—is a numerical method to solve Laplace's equation ($\nabla^2 z = 0$). This equation is the mathematical condition for a minimal surface (zero mean curvature) when the boundary is fixed. Each relaxation step reduces the total area, driving the system toward equilibrium.

Can a real soap film have the complex 3D shape shown here?

Yes, provided the wire frame is rigid and the film is stable. This is the classic 'Plateau's problem' named after the 19th-century physicist Joseph Plateau, who experimentally determined the rules for soap film structures. The simulator's model accurately captures the equilibrium shape for a single, continuous film.

Why are there colorful patterns on the simulated film?

The colors are a schematic representation of thin-film interference. In reality, light waves reflecting off the front and back surfaces of the thin soap film interfere constructively or destructively based on the local film thickness, creating iridescent bands. Here, color is used as a visual proxy, often mapping to the local height or curvature of the computed surface.

What does this simulator simplify or leave out?

The model ignores several real-world effects: gravity (which would cause slight thickening at the bottom), the finite thickness and drainage of the film, variations in surface tension, and the dynamics of how the film forms. It assumes a perfectly elastic, weightless film with constant tension, solving only for the final equilibrium geometry.

Related simulators

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NEW SCHOOL

Thin-Film Interference

▶ Launch Simulator

Wedge fringes: n , d , θ ; $\cos^2(\delta/2)$ colors and $I(\lambda)$ at mid thickness.

NEW SCHOOL

Newton's Rings

▶ Launch Simulator

Spherical lens on flat glass: air-film thickness fringes; optional π phase on reflection.

NEW SCHOOL

Anti-Reflection Coating

▶ Launch Simulator

Air–film–glass at normal incidence: $R(\lambda)$ from two-interface interference; quarter-wave design.

NEW SCHOOL

Thin Lens Equation

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Hyperbola $d_i(d_o)$, pole at $d_o = f$, schematic + $1/d$ check.

SCHOOL

Diffraction

▶ Launch Simulator

Single and double slit with interference patterns.

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Chromatic Aberration

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Cauchy $n(\lambda)$; thin-lens $f(\lambda)$; paraxial rays R/G/B.